3. Create a class named MyMath that performs different mathematical operations. The class must contain a single method named calculate() that can perform different operations depending on the user's input. The method should be able to perform the following operations: Calculate the sum of the first n natural numbers, Display the first n prime numbers, Generate the Fibonacci series for n numbers, Calculate the factorial of a number. The method should accept two parameters: operation name, the value of n. Extend the program so that the user can choose the required operation from the keyboard and display the corresponding result. Write a python program to design a Mathematical utility tool for students where they can select a mathematical operation and obtain results using a single reusable class.

<u>Program:</u>

```
class MyMath:
    def calculate(self, operation, n):
        if operation == "Sum":
            result = n* (n+1)//2
```

```
print ("sum of first ", n, "natural numbers : ", result)
elif operation == "prime":
    num = 2
    count = 0
    print ("First", n, "prime numbers :", end=" ")
    while count < n:
        is_prime = True
        for i in range (2, num):
            if num % i == 0:
                is_prime = False
                break.
            if is_prime:
                print (num, end=" ")
                count += 1
        print( )
elif operation == "fibonacci":
        a, b = 0,
        print ("Fibonacci series:")
        for i in range(n):
            print (a, end=" ")
            a, b = b, a+b
        print()
elif operation == "factorial":
        fact = 1
        for i in range (1, n+1):
            fact *= i
            print ("factorial of", n, "is:", fact)
else:
        print ("Invalid operation")
```

```
math_tool = MyMath()
print("1. Sum of first n natural numbers")
print("2. First n prime numbers")
print("3. Fibonacci series")
print("4. Factorial ")
choice = int(input("Enter your choice (1-4) : "))
n = int(input("Enter the value of n: "))
if choice == 1 :
    math_tool.calculate("sum",n)
elif choice == 2 :
    math_tool.calculate("prime",n)
elif choice == 3 :
    math_tool.calculate("Fibonacci", n)
elif choice == 4 :
    math_tool.calculate("factorial, ",n)
else :
    print("Invalid choice").
```

```
1. Sum of first N natural numbers
2. First N prime numbers
3. Fibonacci series
4. Factorial
Enter your choice (1-4): 1
Enter the value of n: 6
Sum of first 6 natural numbers: 21
PS C:\Users\DELL\OneDrive\Desktop\Python> python -u "c:\Users\DELL\OneDrive\Desktop\Python\conditionalstatements.py\listmethods.py\math.py"
1. Sum of first N natural numbers
2. First N prime numbers
3. Fibonacci series
4. Factorial
Enter your choice (1-4): 2
Enter the value of n: 10
First 10 prime numbers:
2 3 5 7 11 13 17 19 23 29
PS C:\Users\DELL\OneDrive\Desktop\Python> python -u "c:\Users\DELL\OneDrive\Desktop\Python\conditionalstatements.py\listmethods.py\math.py"
1. Sum of first N natural numbers
2. First N prime numbers
3. Fibonacci series
4. Factorial
Enter your choice (1-4): 3
Enter the value of n: 10
Fibonacci series:
0 1 1 2 3 5 8 13 21 34
PS C:\Users\DELL\OneDrive\Desktop\Python> python -u "c:\Users\DELL\OneDrive\Desktop\Python\conditionalstatements.py\listmethods.py\math.py"
1. Sum of first N natural numbers
2. First N prime numbers
3. Fibonacci series
4. Factorial
Enter your choice (1-4): 4
Enter the value of n: 10
Factorial of 10 is: 3628800
```